

AI Interview Simulator - Complete Project Blueprint

Purpose

This document captures the complete vision of the project so that a coding agent or developer can understand the intended product without additional context. The goal is not merely to build another LLM interview chatbot, but an AI-powered interview coach that conducts realistic interviews, adapts to the candidate, continuously evaluates performance, and produces an evidence-based feedback report.

Problem Statement

Existing interview simulators either ask fixed questions, lack contextual follow-up questions, or provide generic scores. This project focuses on realistic interviewer behaviour, adaptive questioning, multimodal analysis (speech, voice, camera), and detailed personalized feedback.

Core Goals

- Realistic technical and HR interviews.
- Adaptive interview difficulty instead of fixed easy/medium/hard.
- Context-aware follow-up questions.
- Free/open-source stack wherever possible.
- Continuous background evaluation during the interview.
- Detailed report that teaches the user how to improve.

User Flow

Homepage → Technical Interview or HR Interview.

Technical Interview → Resume Upload → Role Selection (SDE, AI/ML, Data Scientist/Analyst) → Camera/Mic Check → Fullscreen → Interview → Report.

HR Interview skips resume upload and starts after permissions.

Interview Design

The AI interviewer speaks questions using browser Text-to-Speech. A small transcript of the last question is displayed only as a reference. The candidate answers using the microphone while live speech-to-text runs continuously to minimize waiting time. The user can end the interview early. Target interview duration is approximately 15–20 minutes with adaptive depth rather than a fixed number of questions.

Adaptive Difficulty

There are no visible difficulty levels. After each answer the system decides whether to increase, maintain or decrease complexity. Good answers receive deeper follow-up questions while weaker answers receive simpler questions or topic changes.

Architecture

Each submitted answer triggers two parallel pipelines.

Pipeline A (Real-Time): Speech-to-text completion → LLM understands answer → Generates contextual next question → Returns interviewer response.

Pipeline B (Background): Speech metrics, voice metrics, camera metrics, technical evaluation, communication evaluation, per-question written feedback, scores, and observations are computed and stored immediately. This ensures that the final report requires only summarization instead of reprocessing the entire interview.

Per Question Storage

For every question store: question text, transcript, technical score, communication score, confidence-related metrics, speech metrics (pace, pauses, fillers), camera metrics (eye contact, face visibility), and a ready-to-display feedback paragraph explaining strengths, weaknesses, and improvements for that answer.

Final Report

The report should be narrative rather than score-only. It includes: Interviewer's overall assessment. Performance breakdown. Detailed technical evaluation by topic. Communication analysis. Speech analysis (pace, pauses, filler words). Camera analysis (objective metrics only). Resume consistency analysis. Question-by-question feedback with improvement suggestions. Model answer for weaker responses. Personalized learning plan.

Evaluation Philosophy

Technical interviews prioritize correctness, problem-solving, implementation depth, and resume consistency. HR interviews prioritize communication, confidence, behavioural responses, clarity, and professionalism. The weighting differs, but the reporting format remains consistent.

Technology Stack

Frontend: React + TypeScript + Tailwind.

Backend: FastAPI.

Database: PostgreSQL.

LLM: Groq API.

Speech-to-Text: faster-whisper.

Voice Analysis: librosa.

Camera Analysis: MediaPipe.

Charts: Recharts.

Authentication: JWT.

Future Enhancements

Company-specific interviews, coding interviews, multilingual support, recruiter dashboard, interview history analytics, RAG-based company preparation, and downloadable PDF reports.

Implementation Notes for Coding Agent

Always prioritize responsiveness. Never block the interview while generating detailed analytics. Generate the next question immediately and perform analysis asynchronously. Store user-facing per-question feedback during the interview so the final report only synthesizes stored evaluations into a comprehensive assessment.